

WASP-36b

R-1190

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_210203 · created 2026-07-31T05:38:29+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459248.7468 +/- 0.0041 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.048 +/- 0.049
Transit depth (Rp/Rs)^2	0.23 +/- 0.47 [%]
Orbital Inclination (inc)	88.13 +/- 2.87
Airmass coefficient 1 (a1)	1.5 +/- 0.75
Airmass coefficient 2 (a2)	-0.28 +/- 0.67
Scatter in the residuals of the lightcurve fit is	48.49 %
Best Comparison Star	None
Optimal Aperture	1.11
Optimal Annulus	5.5
Transit Duration (day)	0.0838 +/- 0.009

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.048 ± 0.049 vs 0.1368 (deviation 1.81σ)
Duration fit vs published	0.0838 d vs 0.0758 d (deviation 0.89σ)
Mid-time O–C	-10.1 min
Depth SNR	1.0
Residual scatter	48.49 %

Light curve

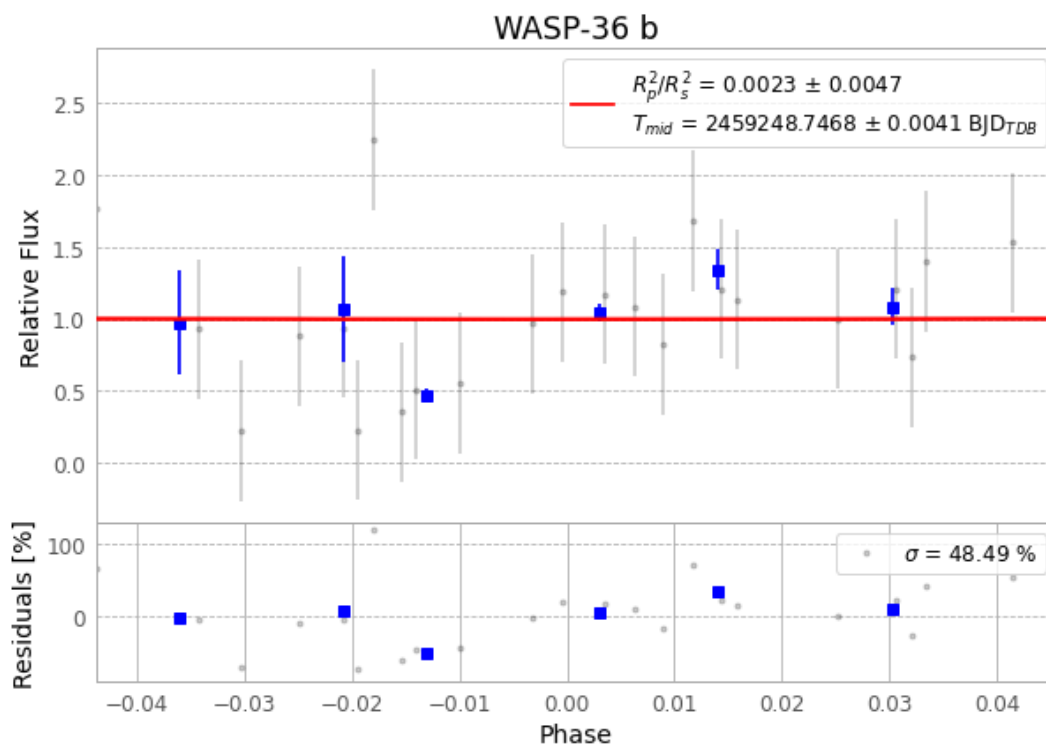


Figure 1 — FinalLightCurve_WASP-36 b_2021-02-02.png (SHA-256 e338e6db92e21b07...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [276, 113], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 961a8da2524caeb7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

69 FITS frames (45 MB), WASP-36210203040612.FITS → WASP-36210203073012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-210203.log (a07aa4308319...), FinalLightCurve_WASP-36 b_2021-02-02.png (e338e6db92e2...), FinalLightCurve_WASP-36 b_2021-02-02.csv (7456bd7fb276...)

Compute resources

Wall time 220.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 05:38Z.